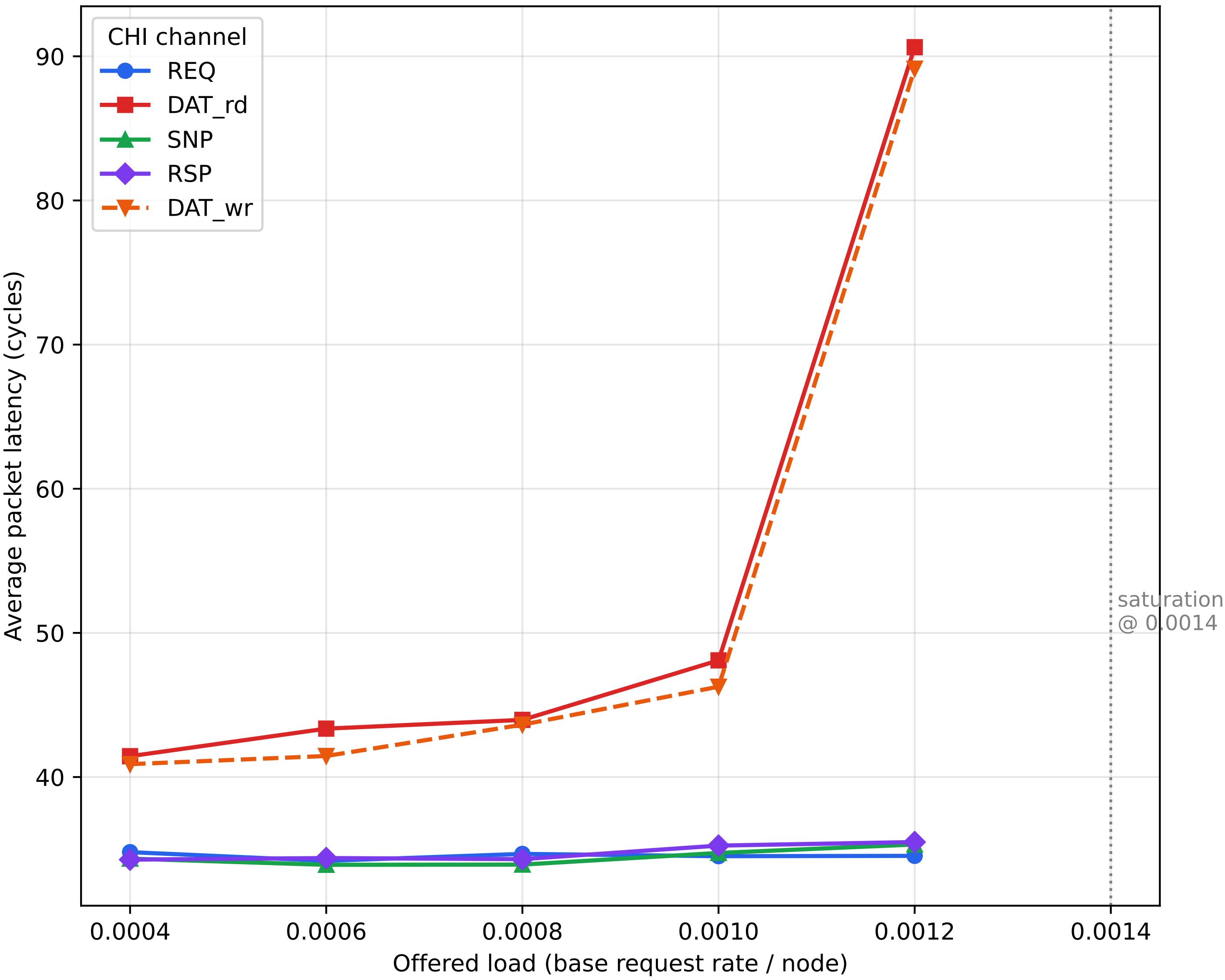
6x7 Mesh + CHI V2 (class_subnet patch)
Per-Channel Latency vs. Offered Load



Key finding: the control channels (REQ / SNP / RSP, 1-flit) stay flat at ~34 cycles, while the two 5-flit DATA flows (DAT_rd HN->RN and DAT_wr RN->RN) share the single DAT subnet and climb steeply -- the DAT physical channel saturates first (knee ~ 0.0012, unstable at 0.0014). This mirrors real CHI, where DAT bandwidth is the system bottleneck.

168 nodes (84 RN + 84 HN) | subnets=4 | 5 classes via class_subnet={0,1,2,3,1} | snoop=1.0, write_fraction=0.3

V2 Channel Mapping & Sweep Data

4 channels = 4 subnets; DAT subnet shared by read+write data

class	CHI channel	subnet	size	direction
c0	REQ	0	1	RN->HN
c1	DAT (read)	1	5	HN->RN
c2	SNP	2	1	->RN
c3	RSP	3	1	->HN
c4	DAT (write)	1	5	RN->HN (NEW vs V1)

Per-channel average packet latency (cycles)

offered	REQ	DAT_rd	SNP	RSP	DAT_wr
0.0004	34.8	41.4	34.3	34.3	40.9
0.0006	34.2	43.4	33.9	34.4	41.5
0.0008	34.7	44.0	33.9	34.3	43.6
0.001	34.5	48.1	34.7	35.2	46.3
0.0012	34.5	90.6	35.3	35.5	89.1
0.0014	SAT	SAT	SAT	SAT	SAT

Highlighted row = last stable point (knee). DAT columns drive saturation.